



CANONICAL PROTOCOL MATERIALS · PUBLIC

The Aethelred Protocol

Sovereign Infrastructure for Verifiable AI Computation

A Layer 1 settlement network in which consensus, confidentiality, and cryptographic evidence are integrated as one engineered surface, built for regulated and high-assurance AI workloads rather than generic blockspace.

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Author	Ramesh Tamilselvan, Founder & Chief Architect
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Native token	AETHEL, 10,000,000,000 fixed at genesis, 0% post-genesis inflation
Target TGE window	Abu Dhabi Finance Week (ADFW), December 2027, subject to audit and operational-readiness gates
Classification	Public, approved for website publication and Registrar review

This whitepaper is a public protocol document prepared for website publication and for review by the appointed Company Service Provider, Foundation legal counsel, and the ADGM Registration Authority pursuant to the Distributed Ledger Technology Foundations Regulations 2023 (“DLTR 2023”). It describes the current technical architecture, governance controls, operating principles, and disclosure posture of the Aethelred protocol. It does not state or imply that any regulatory registration or approval has been completed, that any token sale, listing, or market-maker arrangement has occurred, or that any statement herein is legal or financial advice or an offer of securities or regulated financial products.

AETHELRED DLT FOUNDATION

ABSTRACT**Verifiable AI, settled on-chain**

Aethelred is a sovereign Layer 1 blockchain engineered for regulated and high-assurance AI workloads. Its objective is to let enterprises run sensitive or economically significant AI workflows with cryptographic evidence that legal, security, compliance, and audit stakeholders can independently review.

The network unifies six properties at the protocol layer: deterministic blockchain settlement; Trusted-Execution-Environment (TEE) confidential execution; zero-knowledge machine-learning (zkML) proof verification; portable evidence binding through Digital Seals; policy-aware data and execution routing; and a fixed-supply token economy in which release and disclosure are governed rather than improvised.

The thesis is straightforward. In regulated environments, trust must not depend on vendor assertions alone. Aethelred is built so that participants can prove what ran, where it ran, under which controls it ran, and how the resulting evidence was bound to an independently verifiable record, and so that the protocol's own public commitments remain consistent with the evidence and approvals actually in place at any point in time.

Regulatory framing. The protocol is operated by the Aethelred DLT Foundation, a foundation in the process of registration under DLTR 2023 in the Abu Dhabi Global Market (ADGM). Its DLT Purposes are to use, deploy, develop, facilitate and support distributed ledger technology and to issue the AETHEL token, which is a protocol-utility token and not a token regulated by the ADGM Financial Services Regulatory Authority (FSRA). The Foundation does not conduct, and this document does not offer, any activity requiring a Financial Services Permission.

SECTION 1**Executive summary**

Aethelred is positioned as an AI-native Layer 1, not a generalised blockspace network. The protocol is designed for institutions that need computation, verification, and evidence portability integrated into the settlement layer itself rather than bolted on at the application layer.

1.1 Five public protocol properties

1. **Confidential execution** for sensitive AI workloads via attested TEE backends.
2. **Independently checkable verification** through governed attestation and proof-system pathways.
3. **Portable audit evidence** through Digital Seals as first-class settlement artifacts.
4. **Utility-first, fixed-supply token economics** with no post-genesis inflation under the canonical posture.
5. **Disciplined public disclosure** aligned to legal review, a claims-register, and benchmark gates.

1.2 Document at a glance

TOPIC	CURRENT PUBLIC POSITION
Protocol type	AI-native sovereign Layer 1 blockchain

Operating entity	Aethelred DLT Foundation (in formation, ADGM) under DLTR 2023
Native token	AETHEL (protocol-utility token; not an FSRA-regulated token)
Supply model	Fixed supply of 10,000,000,000 AETHEL minted at genesis
Post-genesis inflation	0%
Consensus primitive	Proof-of-Useful-Work (PoUW) coupled to verified AI computation
Verification model	TEE attestation paired with zero-knowledge proof verification
Evidence model	Digital Seals as portable, externally reviewable evidence objects
Cryptographic posture	ML-DSA signature support and ML-KEM-768 default transport profile
Public performance rule	Benchmark-gated; no unverified throughput claims published
Target TGE window	ADFW, December 2027, gated on audit and operational readiness

1.3 Why the problem matters

Large enterprises increasingly deploy AI in decisions that create legal, capital, or public-safety exposure. In these settings the question is not only whether a model produced an output, but whether that output can be trusted, reproduced, and defended under scrutiny. Most production deployments still rely on fragmented evidence spread across cloud logs, application telemetry, and internal controls. That is rarely sufficient for regulated production systems where decisions must be defensible to auditors, regulators, courts, and counterparties.

1.4 What Aethelred contributes

Aethelred's contribution is the union of compute, verification, and evidence into a single protocol surface. Verification is not an afterthought layered on application output; it is a first-class settlement condition. Workloads that fail verification fail closed and do not produce a Digital Seal. Workloads that succeed are bound by the seal to the validator that executed them, the environment they ran in, and the on-chain state that recorded them. This architecture replaces narrative trust with protocol-verifiable evidence.

SECTION 2

Problem statement & design principles

PROBLEM	MANIFESTATION	PROTOCOL RESPONSE
Execution trust	Buyers and auditors cannot independently prove that the claimed model, environment, and output correspond to the computation that actually occurred.	TEE attestation and zkML proof verification, bound by Digital Seals.
Confidentiality & sovereignty	Sensitive workloads require controls over where data runs, how it is isolated, and which operators may access it.	Policy-aware lane scheduling and jurisdictional routing under governance control.

Evidence portability	Even where attestation exists, evidence is fragmented across cloud systems and logs rather than exposed as durable, portable artifacts.	Digital Seals as standardised, externally reviewable settlement objects.
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2.1 Design principles

PRINCIPLE	MEANING IN PRACTICE
Confidentiality with auditability	Sensitive inputs may remain protected while outcome evidence remains independently reviewable.
Fail-closed operation	Production paths reject incomplete, simulated, or inconsistent evidence rather than silently degrading.
Evidence portability	Results are exportable as reusable audit artifacts, not only as internal logs.
Governance over optimism	Public claims follow evidence and approvals, not roadmap ambition.
Modular assurance	Compute, proof, attestation, and policy layers can evolve without breaking the trust model.
Demand-gated incentives	Validator and ecosystem rewards track verified useful work, not calendar inflation.
Sustainable compute	Consensus energy performs real AI inference; the network does not burn energy on hash puzzles or idle stake.

2.2 Scope of the design goal

The design goal is not to be a generic high-throughput chain. It is to be the most defensible platform for regulated enterprises to run AI with mandatory cryptographic evidence. That goal produces hard constraints: confidentiality must be compatible with auditability; evidence must be portable and machine-verifiable; production paths must fail closed rather than degrade to simulation; performance claims must be benchmark-governed; and legal and commercial claims must follow disclosure state, not pipeline optimism.

SECTION 3

System architecture & network overview

Aethelred is built as a protocol stack with five interacting layers that together form an evidence-aware compute network. Each layer has a distinct responsibility, but the layers are designed so that evidence integrity is preserved end-to-end, from job submission, through execution and verification, into settled state and exportable Digital Seals.

LAYER	RESPONSIBILITY	SURFACE
5 · Disclosure & governance control plane	Claims register, legal review, counterparty state, drift controls.	Public surface
4 · Developer & operator surfaces	Go, Rust, Python, TypeScript SDKs, CLI tooling, APIs, runbooks.	Integration surface

3 · Evidence & sealing layer	Digital Seals, audit surfaces, cross-domain evidence relay.	Evidence binding
2 · Execution & verification layer	TEE attestation (SGX, Nitro, SEV-SNP, Azure CVM, GCP CVM) plus zkML proof verification.	Confidential compute
1 · Consensus & state settlement	Proof-of-Useful-Work, deterministic finality, governance recording.	Settlement base

3.1 Core protocol components

COMPONENT	ROLE
Consensus engine	Orders transactions, records governance actions, and finalises valid state transitions.
PoUW scheduler	Assigns and gates useful-work execution within protocol rules and lane policies.
Verification layer	Checks attestation and proof artifacts against accepted formats, registries, and policies.
Seal layer	Issues portable evidence objects tied to settled state and supports cross-domain export.
Integration layer	Exposes SDKs, APIs, and operational surfaces for developers and institutional integrators.

3.2 Node and operator classes

OPERATOR CLASS	PRIMARY RESPONSIBILITY	PUBLIC POSTURE
Validators	Settlement, proposal participation, evidence-aware execution.	Production and testnet roles differ by release bundle.
Sentry & edge nodes	Network hardening and topology isolation.	Operational detail governed by runbooks.
Proof & verification services	Proof validation, attestation handling, evidence normalisation.	Must remain consistent with canonical verifier support.
Enterprise integrators	API and SDK consumption, workflow integration.	Supported through governed integration surfaces.

SECTION 4

Consensus & verified compute

Aethelred uses a Proof-of-Useful-Work (PoUW) design in which consensus and verified AI computation are linked. Validators do not earn block-production credit by burning energy on cryptographic puzzles or by passively locking capital. They earn it by executing economically meaningful AI work whose correctness is independently verifiable through attestation and proof artifacts.

At a high level: jobs are routed to the appropriate execution lane; execution occurs inside an approved confidential-compute environment; the governed verification path produces attestation and proof

artifacts; evidence is checked against protocol rules; and successful results are sealed and settled on-chain. Failures at any stage halt the pipeline rather than degrading silently to a permissive output.

4.1 Lanes and workload separation

The protocol scheduler separates workloads into dedicated lanes so that large proof-heavy jobs do not block smaller or faster workloads. Lanes are an architectural control, not a public throughput claim: a fast inference lane for small-model inference and short-context retrieval; a standard enterprise lane for medium scoring, classification, and decisioning; and a heavy compute lane for proof-heavy or large-model jobs requiring extended attestation.

4.2 Proof of Useful Work

PoUW is the foundational consensus primitive that distinguishes Aethelred from both traditional Proof-of-Work and Proof-of-Stake architectures. In conventional PoW systems, validators expend computational resources on cryptographic puzzles that serve no purpose beyond block production. In PoS systems, validators lock capital but the consensus mechanism does not directly connect to the economic output of the network. PoUW requires that block-producing validators perform economically meaningful AI computation as the basis of their consensus participation. The core principles are:

- **Work must be useful.** Every unit of computation that contributes to consensus must also produce a verifiable, economically relevant output: an AI inference result, a model evaluation, a data transformation. Idle or synthetic work is rejected.
- **Work must be verified.** Each useful-work output passes through the protocol's dual verification path. Consensus credit is awarded only when both execution and evidence meet protocol standards.
- **Work must be governed.** The scheduler assigns work based on declared capabilities, hardware attestation, jurisdictional compliance, and lane capacity. Validators cannot self-select high-reward jobs or replay previous outputs.
- **Work must be accountable.** Every PoUW cycle produces a Digital Seal binding validator identity, workload identity, execution evidence, and verification artifacts into a portable, auditable record. Outputs that fail verification are subject to slashing.

Sustainable by design. Because consensus credit is earned by performing real AI inference, network security and economic utility are the same activity. Unlike Proof-of-Work, there is no wasted hashing; unlike Proof-of-Stake, the consensus mechanism is directly coupled to the network's core output. Energy spent securing Aethelred is energy spent on computation a customer actually needs.

4.3 Anti-gaming and quality controls

Any compute-linked consensus model must account for strategic behaviour. The public architecture emphasises workload classification with policy-aware lane routing; proof and attestation validation against canonical registries; domain binding and replay resistance on every seal; slashing and accountability tied to evidence quality; and evidence-based rejection where job integrity or provenance is uncertain.

SECTION 5

Verification model

The core trust model combines Trusted Execution Environments for confidential execution and measurement, and zero-knowledge proof systems for independently checkable verification. Neither path alone is sufficient for the regulated workloads Aethelred targets. TEE attestation establishes that a specified binary ran in a measured environment; zkML proves correctness of computation without revealing the underlying data or weights. The hybrid surface unifies both into a single verifier interface.

5.1 Enterprise hybrid path

The enterprise trust posture is centred on a governed hybrid path. Execution occurs inside an approved TEE backend; the corresponding proof artifact is checked through the supported proof-verification surface; and mismatches or incomplete evidence fail closed. This is the highest-assurance path for regulated workloads. Workloads that opt out of one of the two halves are routed to lanes with explicitly lower public assurance and documented as such in seal metadata.

5.2 TEE and proof-system coverage

The current public architecture describes support for multiple confidential-compute backends, including Intel SGX, AWS Nitro, AMD SEV-SNP, Azure Confidential VMs, Google Confidential VMs, and NVIDIA confidential-computing paths where applicable. The protocol surface supports multiple proof-system paths through a unified verification interface. Public materials describe support posture and controls qualitatively and do not present unverified benchmark-superiority or proof-speed claims.

5.3 Failure handling and evidence rejection

A verification system loses credibility if incomplete evidence is silently accepted. Incomplete attestation, malformed proof artifacts, or inconsistent execution evidence must not be normalised into green outcomes. Any of these conditions causes the seal to be withheld and the verification record to be marked as a rejected event, with explicit operator-facing diagnostics.

SECTION 6

Digital Seals

Digital Seals are the protocol's portable evidence artifact and the central output of the verified compute pipeline. A seal is what an auditor, regulator, counterparty, or downstream system consumes to confirm that a given AI workload executed under the controls Aethelred describes.

6.1 What a seal binds

- **Workload identity:** the canonical job descriptor.
- **Model or artifact identity:** the executed binary or model commitment.
- **Input and output commitments:** content-addressed references allowing reproducibility without exposing data.
- **Execution evidence:** TEE attestation, environment measurement, and operator identity.
- **Verification evidence:** proof artifact reference and verifier registration metadata.
- **Settlement context:** block height, finality status, and governance state at sealing time.

6.2 Why seals matter

Most enterprise systems can produce logs. Fewer can produce a portable, externally reviewable evidence object that moves across operators, auditors, counterparties, and systems of record. Digital Seals are designed to be the primary artifact reviewed during a compliance audit, the primary record cited in a counterparty dispute, and the primary input for downstream automation that depends on verifiable provenance. The verification tools that allow Tokenholders and third parties to check seal information independently are part of the published DLT Framework.

SECTION 7

Execution environment

Aethelred is designed around an AI-native execution posture rather than generic smart-contract execution alone. The compute posture is centred on model execution, verification, and evidence generation, not only on transferring tokens or recording state mutations. Public materials describe an execution surface with AI-oriented precompiles; system contracts for job and seal lifecycle handling; confidential-compute backends for execution; proof-verification surfaces for high-assurance settlement; and SDKs and APIs for integration. These modules are governed through the same release-bundle and disclosure controls as the consensus engine itself.

SECTION 8

Data, privacy & vector workloads

The protocol is intended for sensitive and regulated data flows. That requires policy-aware treatment of workloads, evidence of the environment in which data was handled, explicit boundaries between public and confidential state, and careful treatment of vector and AI retrieval layers.

8.1 Sovereign data model

Aethelred supports jurisdiction-aware and policy-aware workload treatment. Data policy and execution policy are treated as protocol concerns, not only application concerns: routing decisions, attestation requirements, and evidence binding are all influenced by the policy tags attached to a workload. This directly supports the data-protection and data-security Relevant Obligations to which the Foundation is subject under DLTR 2023.

8.2 Vector Vault

A verified Vector Vault data plane anchors namespace metadata and committed vector snapshots on-chain while production embedding and approximate-nearest-neighbour backends run behind attested execution paths. This preserves auditability without forcing a full production vector database into consensus state.

SECTION 9

Post-quantum & cryptographic posture

Aethelred uses a hybrid cryptographic posture rather than relying on a single primitive: ML-DSA-based post-quantum signature support together with classical compatibility where required; ML-KEM-768 as the current default transport profile; higher-security transport profiles available for future governance activation; and fail-closed production rules rather than soft simulation defaults. This document does not claim a completed migration away from all classical dependencies across every integration surface;

it states the current governed posture and identifies which profiles are available for future governance activation as standards mature. A description of the cryptographic technological tools used is maintained in the published DLT Framework, as required by DLTR 2023.

SECTION 10

Security model

The security model depends on more than cryptography. It includes attestation and measurement governance; replay resistance and domain binding; verifier registration; production-mode rejection of simulated or incomplete evidence; operator and release controls; and disclosure discipline around what is truly live.

TRUST BOUNDARY	WHY IT MATTERS
Validator boundary	Determines who can propose, verify, and settle state.
TEE boundary	Determines the confidentiality and measurement trust surface.
Proof-verification boundary	Determines which claims can be independently checked.
Disclosure boundary	Determines which claims may be published publicly.
Governance boundary	Determines who can change parameters, code, or release state.

Public security language must match real production rules. If a surface is not production-ready, the documentation says so or withholds the claim. Production paths reject incomplete attestation, malformed proof artifacts, or inconsistent execution evidence rather than promoting them to acceptance.

SECTION 11

Audit & assurance programme

Aethelred operates a structured assurance programme so that consensus, execution, verification, wallet, and validator surfaces are independently reviewed before exposure to mainnet conditions. The programme is funded as part of the launch budget, executed through external audit firms, and gated against operational milestones. No surface advances without remediated findings on record. Under DLTR 2023, the Foundation will additionally conduct a Security Audit of the DLT Framework at least once per calendar year and file it with the Registrar within 14 days of completion.

WORKSTREAM	TARGET WINDOW	GATE BEFORE PROCEEDING
Protocol specification & threat-model review	Q3 2026	Before external devnet
Primary protocol audit (consensus & execution)	Q1 2027	Before incentivised testnet
Secondary protocol audit (fixes & high-risk areas)	Q2 2027	Before mainnet candidate

Wallet & smart-contract audit	Q2 2027	Before airdrop claims
Validator & infrastructure review	Q3 2027	Before mainnet launch
Formal verification (critical invariants)	Q3 2027	Before TGE
Bug bounty programme	Q3 2027 onward	Before public mainnet
Incident response & tabletop exercises	Q3 2027	Before exchange readiness
DLT Framework Security Audit (annual, DLTR 2023)	Q3 2027, then annually	Filed with Registrar within 14 days

Critical and high-severity findings must be remediated before the corresponding gate opens. Public materials may not describe a surface as “audited” until the corresponding audit report and remediation tracker are complete and approved through the claims register. The protocol maintains a Security & Slashing Insurance Reserve (2% of total supply) sized for validator-slashing events, verified exploit response, and emergency recovery; it is framed as security capacity, not a price-support or buyback instrument.

SECTION 12

Governance & the DLT Framework

Aethelred is governed through the Aethelred DLT Foundation under DLTR 2023. The Foundation’s constitutional document is its Charter; its technical governance is published as a DLT Framework; and its economic model is published as the AETHEL Tokenomics Paper. All three are approved by the Foundation Council and are amended only in accordance with DLTR 2023.

12.1 Organisational and governance structure

BODY	ROLE
Founder	Establishes the Foundation and holds reserved powers until Founder Resignation.
Foundation Council (2-16 councillors)	Ensures compliance with the Relevant Obligations and the Charter; holds non-waivable Veto Rights over decisions that would breach the Regulations, public order, the Charter, or the principles of the Whitepaper, or that could harm the DLT Framework.
Tokenholders	Exercise voting rights on defined Delegated Matters (including Qualified Matters such as changes to the Charter, DLT Framework, objects, name, or jurisdiction) through the Voting of Tokenholders.
Guardian	Appointed where the Charter requires or upon Founder Resignation; supervises compliance with the Charter.

12.2 Governance layers

LAYER	PUBLIC DESCRIPTION
Protocol governance	Parameter and change approval within defined bounds.
Release governance	Release-bundle control, branch protection, deployment discipline.
Disclosure governance	Claims register, legal review, counterparty-state control.
Incident governance	Emergency handling, rollback, communications discipline.

12.3 The DLT Framework

The DLT Framework is a separately published document, maintained by the Foundation and available through a public hyperlink, that defines the roles and responsibilities of the Foundation and its Third Party Providers and establishes a comprehensive framework to manage DLT risk, including regular testing and coding controls, post-deployment monitoring, a risk control matrix, a description of the cryptographic tools used, and tools that allow Tokenholders to verify information from the framework. It is confirmed by an independent Security Audit and security-and-compliance report acceptable to the Registrar.

SECTION 13

Token model summary

The native token of the protocol is AETHEL. The full economic specification, including allocation, vesting, unlock guardrails, fee architecture, burn mechanics, and disclosure controls, is set out in the canonical companion document, the AETHEL Tokenomics Paper. The summary below records the public token facts on which this whitepaper relies.

METRIC	CURRENT PUBLIC POSITION
Token	AETHEL
Legal nature	Protocol-utility token; not a token regulated by the FSRA (DLTR 2023 s.3(1)(b))
Total supply	10,000,000,000 AETHEL, fixed at genesis
Post-genesis inflation	0%
Initial circulating supply at TGE	1,000,000,000 AETHEL (10.0% of total supply)
Monthly unlock ceiling	1.25% of total supply, additionally throttled by liquidity-volume gates
Denominations	uaethel (6 decimals); 18-decimal execution compatibility where required
Target TGE window	ADFW, December 2027, gated on audit and operational readiness

AETHEL has roles in staking and validator security, slashing collateral, fee settlement, governance participation, verified-compute settlement, and burn-based supply reduction. It is explicitly not

described as an ownership interest, a claim on revenue, a debt instrument, a redemption right, a guarantee of appreciation, or a promise of exchange access.

SECTION 14

Interoperability & settlement

Aethelred is designed to participate in wider enterprise and multi-chain workflows. The public architecture references bridge contracts, proof relays, seal verification across domains, and governance-controlled emergency mechanisms, all governed through the same release controls and security gates as core protocol modules. Seals are intended to travel with assets and decisions across domain boundaries without losing evidence integrity. Interoperability must not weaken the proof or disclosure model: cross-system interoperability remains governed by verification and operational controls, not by the lowest-assurance counterparty in the chain.

SECTION 15

Benchmark & claims discipline

The project maintains a benchmark claims register. The governing rule is that every public performance number must have a reviewed and verified benchmark path before publication.

CLAIM TYPE	PUBLICATION RULE
Architecture claim	Allowed if accurately reflected in code and docs.
Security claim	Allowed only if consistent with real controls.
Performance claim	Allowed only from reviewed benchmark packs.
Commercial claim	Allowed only after approval and executed status where relevant.
Regulatory claim	Allowed only if evidenced and legally approved.

SECTION 16

Developer platform

The developer surface includes Go, Rust, Python, and TypeScript SDKs, CLI tooling, APIs, and local or hosted environments for both protocol developers and enterprise integrators. Public developer materials distinguish clearly between local simulation or dev-only paths, hosted testnet paths, and production-grade operator paths, so that examples never silently depend on simulated or insecure fallback behaviour.

SECTION 17

Testnet & operational readiness

The public testnet posture is an operational-readiness programme rather than a marketing claim. The project advances through named phases; each has explicit validator-count, uptime, and PoUW-job-volume targets; and each exits only when its release gate is satisfied. None of the figures below are throughput claims; they are operational targets used to gate progression.

PHASE	WINDOW	VALIDATORS	UPTIME	POUW JOBS / MO
Internal devnet	Q3 2026	20	95%+	1,000
Public testnet	Q4 2026	150	98%+	25,000
Incentivised testnet	Q2 2027	500	99%+	100,000
Mainnet candidate	Q3 2027	1,000	99.5%+	250,000
Mainnet & TGE	ADFW Dec 2027	n/a	n/a	n/a

Each phase includes a documented exit criterion. A phase is complete not because a date passes but because its release gate is satisfied. The mainnet candidate and TGE targets are contingent on satisfaction of the audit gates in Section 11.

SECTION 18

Roadmap & sequencing

The roadmap runs from 2026 to the ADFW December 2027 TGE across parallel workstreams (foundation & legal, protocol architecture, PoUW, hiring, testnet, security & audit, exchange readiness). Technical readiness, not commercial pressure, drives launch sequencing. Audits, formal verification, and incentivised-testnet performance complete before exchange-readiness or TGE work begins in earnest. Publicly visible milestones are treated as work-in-progress targets; the actual transitions are gated by Sections 11, 15, and 17 and reflected in the claims register before any public statement is updated.

STAGE	WINDOW	HEADLINE
Foundation year	2026	ADGM DLT Foundation registration, public testnet, pre-seed close, first design-partner pilots.
Audit & hardening	H1 2027	Primary and secondary audits, incentivised testnet, seed round.
Mainnet candidate	H2 2027	Formal verification, validator review, mainnet candidate, exchange readiness.
Network launch & TGE	ADFW Dec 2027	Mainnet launch and Token Generation Event, gated on audit and operational readiness.

SECTION 19

Institutional use cases & protocol fit

The following are domains where the protocol architecture is structurally suited, not commercial commitments.

19.1 Financial services

Confidential models, sanctions-aware routing, verifiable execution records, and governed evidence export, including credit and underwriting models, market-abuse surveillance, and model-risk reporting where output justification is itself a regulated artifact.

19.2 Healthcare and life sciences

Jurisdiction-aware handling, confidential execution, and durable provenance over AI-assisted outputs: clinical decision support, medical-imaging triage, pharmacovigilance, and trial-data analytics where consent, location, and audit context must travel with the result.

19.3 Research, mobility, and supply chains

Provenance, evidence portability, and reproducibility for safety-critical decisioning, fleet coordination, and route optimisation where tamper-proof evidence trails support regulatory compliance and liability attribution.

19.4 AI agents

Autonomous agents operating across organisational boundaries require verifiable execution records to establish counterparty trust. Aethelred provides the evidence infrastructure that lets agents prove what actions they took, under which policies, and with what outputs, supporting agent-to-agent commerce and multi-agent orchestration.

SECTION 20

Competitive positioning

DIMENSION	AETHELRED	GENERALISED L1	CENTRALISED AI API
Verifiable AI execution	Core design objective	Usually application-layer	Vendor-assertion based
Confidential compute posture	Native architectural priority	Varies by application	Vendor-controlled
Portable audit evidence	Digital Seals	Fragmented or custom	Usually internal only
Disclosure discipline	Governed claims model	Rarely explicit	Opaque
Regulated-workload fit	Primary focus	Secondary or app-specific	Depends on vendor controls

The competitive frame describes architectural priorities, not measured performance. Comparative performance claims belong in benchmark packs and not in this whitepaper unless verified and approved for release.

SECTION 21

Regulatory & legal posture

The protocol is operated by the Aethelred DLT Foundation, in the process of registration as a DLT Foundation under DLTR 2023 in the ADGM. The Foundation's objects are consistent with the DLT Purposes; it does not conduct any activity for which a Financial Services Permission is required and does not operate as a Non-Profit Organisation. AETHEL is a protocol-utility token and is not a token regulated by the FSRA.

Public wording currently permitted: the project is structured for governed legal and disclosure requirements; ADGM DLT Foundation registration is in preparation; regulatory and legal publication materials remain in preparation.

Public wording not permitted without evidence: any statement of completed regulatory registration, approval, or filing, or any equivalent wording implying completed registration or approval. Any activity requiring regulatory authorisation will be undertaken only with the appropriate approval structure in place.

SECTION 22

Public disclosure boundary

DISCLOSURE CLASS	CURRENT STATE
Architecture and protocol design	Public
Code-backed token posture	Public
Allocation and vesting structure	Public (in the Tokenomics Paper)
Roadmap and operational targets	Public, subject to gates
Performance metrics	Benchmark-gated
Counterparty naming	Executed-only and approval-gated
Regulatory status claims	Evidence- and legal-approval-gated
Listing, market-maker, and pricing details	Withheld pending approval

SECTION 23

Risk factors

Key public risk categories include technical execution risk; benchmark and infrastructure readiness risk; developer-path maturity risk; legal and regulatory timing risk (including ADGM registration timing); counterparty execution risk; launch-timing and disclosure-timing risk; and adoption risk. The ADFW December 2027 TGE target compresses the audit and assurance sequence into 2027 and is contingent on each gate in Section 11 being satisfied. No reader should treat this whitepaper as a guarantee of launch sequence, market outcome, or regulatory result. Aethelred is a governed protocol programme with live technical artifacts, with certain legal, commercial, and launch disclosures intentionally withheld until evidence and approvals are complete.

SECTION 24

Conclusion

Aethelred is built around a practical thesis: regulated AI requires stronger evidence than vendor trust alone. The protocol combines deterministic settlement, confidential execution, proof verification, portable evidence in Digital Seals, fixed-supply token discipline, and governed disclosure controls,

operated by a DLT Foundation under a recognised legal framework. By the time AETHEL is launched on mainnet, the evidence, the audits, the testnet performance, and the legal posture should each be capable of withstanding the scrutiny that regulated counterparties bring. The roadmap, the assurance programme, and the disclosure discipline are designed to make that claim defensible, not merely aspirational.

APPENDIX A

Public statements that must remain true

- External performance numbers publish only from verified benchmark packs and reviewed claims-register entries.
- The network is described with a fixed supply of 10 billion AETHEL tokens and 0% post-genesis inflation.
- Initial circulating supply at TGE is 10.0% of total supply (1,000,000,000 AETHEL).
- Monthly scheduled releases are capped at 1.25% of total supply and additionally throttled by trailing-volume and free-float gates.
- ML-KEM-768 is the current default transport profile.
- AETHEL is a protocol-utility token and not a token regulated by the FSRA.
- Counterparty names remain withheld until executed and approved for disclosure.
- Public regulatory wording remains limited to the current preparation-stage posture; ADGM DLT Foundation registration is described as in preparation, not complete.

Appendix B · Glossary

TERM	MEANING
Digital Seal	Portable evidence artifact linking execution, verification, and settlement context.
PoUW	Proof-of-Useful-Work design linking useful compute and protocol settlement.
TEE	Trusted Execution Environment, a hardware-isolated compute domain with measurement support.
zkML	Zero-knowledge proof systems applied to ML workflows or outputs.
DLT Framework	The Foundation's published technical-governance and risk framework under DLTR 2023.
Vector Vault	Verified vector data plane anchoring committed snapshots on-chain.

Appendix C · Reference documents

DOCUMENT	PURPOSE
AETHEL Tokenomics Paper	Canonical public tokenomics and economic-design document.

Aethelred DLT Foundation Charter	Constitutional document of the Foundation under DLTR 2023.
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Aethelred DLT Framework	Published technical-governance and risk-management framework.
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